Shuo Sha

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EDUCATION

Columbia University, *Fu Foundation School of Engineering and Applied Science* | New York, NY

- Bachelor of Science in Applied Mathematics, GPA 3.97
- Minors: Computer Science, Philosophy
- Expected Graduation: May 2026
- Relevant Coursework: Machine Learning, Robot Learning, Deep Learning for Robotics, Modern Analysis, Probability Theory

RESEARCH EXPERIENCES

Columbia University & Scenix Inc (Advisor: Prof. Yunzhu Li)

Real-to-Sim Policy Evaluation with Gaussian Splatting Simulation | Jun. 2025 – Sep. 2025

- Developed a real-to-sim policy evaluation framework that constructs soft-body digital twins from real-world videos using 3D Gaussian Splatting.
- Benchmarked four imitation learning policies (Diffusion Policy, ACT, Pi-0, SmolVLA) on deformable manipulation tasks, demonstrating strong sim and real rollout correlation.
- Under Review at *ICRA 2026*. **Co-first author**
- Website: https://robopil.github.io/real-to-sim-policy-eval/

Columbia University – RoboPIL Lab (Advisor: Profs. Yunzhu Li & Antonio Loquercio)

Residual Policy for Fine-Grained Manipulation | Sep. 2024 – Present

- Designed a residual-augmented teleoperation framework where human operators provide high-level guidance and a residual policy ensures low-level robustness.
- Preparing deployment on NIST contact-rich assembly benchmarks to improve efficiency of human data collection.
- In Preparation for *CVPR 2026*. **First author**

Columbia University – A2R Lab (Advisor: Prof. Brian Plancher)

Kaczmarz Algorithm for Online Inertial Parameter Estimation | Sep. 2023 – Dec. 2024

- Extended the Kaczmarz method with greedy row selection and Polyak averaging for fast, stable inertial parameter adaptation at low per-iteration cost.
- Evaluated on quadrotor tracking with unknown payloads and a synthetic embedded benchmark, achieving significant speed-ups and improved end-to-end performance.
- Under Review at *ICRA 2026*. **First author**
- Link: https://arxiv.org/abs/2510.04839

Beijing Academy of Artificial Intelligence (Advisor: Prof. Zongqing Lu)

VLM-Guided Locomotion for Agile Humanoid Behaviors | Jun. 2024 – Sep. 2024

- Trained vision-based locomotion policies for the Unitree H1 humanoid robot on agile tasks (leaping, squat-walking, hurdling).

- Distilled expert policies into a transformer-based generalist, enabling a video-language model (VLM) to select task-specific policies.

Beijing International Studies University (Advisor: Hua Zhu)

Well-posedness Analysis of 2D Maxwell's Equations | Jun. 2022 – Mar. 2023

- Analyzed the well-posedness of time-harmonic 2D Maxwell's equations that model the Transverse Magnetic Problem using variational formulation and constructed an internal approximation using the finite element method
- Published at *Journal of Mathematics Research*. **First author**

Publications

- **Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions**
  Kaifeng Zhang*, **Shuo Sha*,** Hanxiao Jiang, Matt Loper, Jay Song, Zhuo Xu, Xiaochen Hu, Changxi Zheng, Yunzhu Li. Submitted to *IEEE International Conference on Robotics and Automation (ICRA 2026).*
- **TAG-K: Tail-Averaged Greedy Kaczmarz for Computationally Efficient and Performant Online Inertial Parameter Estimation**
  Shuo Sha, Anupam Bhakta, Zhenyuan Jiang, Kevin Qiu, Ishaan Mahajan, Gabriel Bravo, Brian Plancher. Submitted to *IEEE International Conference on Robotics and Automation (ICRA 2026)*.
- **REFINE: Residual Augmented Teleoperation for FINE-grained Manipulation**
  Shuo Sha, Yixuan Wang, Binghao Huang, Antonio Loquercio, Yunzhu Li. In preparation for *IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2026).*
- **Analysis of 2D Maxwell's Equations in a Time-Harmonic Regime**
  Shuo Sha. Submitted to *Journal of Mathematics Research*, Canadian Center of Science and Education, Vol. 15, No. 2 (2023), https://doi.org/10.5539/jmr.v15n2p1.

Awards

***Semi-finalist**, S.-T Yau High School Science Award* | Jan. 2023

- Regional Second Prize (Top 8 of 5,800+ teams across 1,200 schools), global competition sponsored by Prof. Shing-Tung Yau.

SKILLS

Programming: Python, Java, MATLAB
Frameworks: PyTorch, Isaac Sim/Lab, LeRobot
Robots & Hardware: Unitree H1/G1/GO2, Xarm, ALOHA